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Poster: Automated SIEM Detection Rule Translation System

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***Abstract*—Security Operations Centers (SOCs) depend on SIEM detection rules to identify malicious activity, but detection logic is tightly bound to platform-specific query languages such as SPL (Splunk), KQL (Microsoft Sentinel), and YARA-L (Chronicle). This coupling complicates SIEM migration and multi-platform detection engineering, increasing cost, slowing adaptation, and raising the risk of errors. Detection content becomes fragmented, difficult to standardize or reuse, and prone to semantic inconsistencies and incorrect field mappings, which can lead to detection gaps and false negatives. These issues are intensified by the growing volume and diversity of security telemetry, which increases rule complexity and limits the feasibility of manual cross-platform alignment. Although Large Language Models (LLMs) show promise for automated query translation, unguided generation often produces invalid syntax or semantically incorrect rules, which is unacceptable in security-critical contexts. This paper presents an automated architecture for cross-platform SIEM rule translation that integrates Retrieval-Augmented Generation, deterministic workflow orchestration, and SIEM-native validation to enable stable, accurate, and scalable end-to-end detection rule translation.**

***Keywords*—Large language models (LLMs), Retrieval-Augmented Generation (RAG), SIEM rule translation, Detection Engineering, Security Information and Event Management (SIEM), Security Operations Center (SOC), Workflow orchestration, Chronicle Security**

I. Introduction

Security Operations Centers (SOCs) rely on detection rules to identify malicious activity across enterprise environments. These rules are implemented within Security Information and Event Management (SIEM) platforms such as Splunk, Microsoft Sentinel, and Google SecOps. Each platform uses a distinct query language and data model, including SPL, KQL, and YARA-L, which makes migrating detection logic between SIEM platforms during platform transitions complex and labor-intensive.

As organizations migrate between SIEM platforms, detection engineering teams are often required to manually rewrite and validate large volumes of rules. This process is error-prone, difficult to scale, and highly dependent on platform-specific expertise. Even minor syntactic or semantic differences between query languages can result in incorrect detections or missed security events.

Recent advances in large language models (LLMs) have shown promise in automating code and query translation tasks. However, applying LLMs directly to security detection rules presents significant challenges. Ungrounded generation may introduce hallucinations, incorrect field mappings, or invalid syntax, which is unacceptable in security-critical environments. A survey of existing tools and research prototypes shows that in practice, the quality and operational reliability of automatically translated detection rules often fall below the level expected by production SOC environments.

To address these limitations, this work proposes an automated architecture for a SIEM detection rule translation that combines Retrieval-Augmented Generation [1] with deterministic workflow orchestration and syntax validation. The RAG system is built on LangGraph with hybrid search (vector + BM25), re-ranking via a cross-encoder, syntax validation, and iterative error correction. The system translates security rules between SPL/KQL/YARA-L, generating reasoning steps and confidence scores, demonstrating improved translation quality and consistency.

Early experiments relying on vector similarity demonstrated improvements in translation quality compared to native LLM-based generation, while remaining insufficient for production use. To further enhance retrieval accuracy, we also evaluated BM25-based lexical retrieval, which better prioritized exact field names, operators, and platform-specific syntax. While this improved translation precision and consistency, syntactic errors still occurred in some cases, motivating the adoption of a hybrid retrieval strategy combined with SIEM-native syntax validation and an iterative auto-correction loop to balance semantic coverage and operational accuracy. Together, these components significantly reduced manual analyst rework, improved translation stability across SPL, KQL, and YARA-L, and enabled reliable, end-to-end rule translation suitable for production SOC workflows.

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II. Motivation Example

In Fig. 1 the upper section illustrates the traditional SIEM rule translation workflow. In a conventional process, once a security analyst is required to migrate or replicate a detection rule across SIEM platforms, they typically spend significant time manually analyzing the source rule logic, studying the target query language, and identifying appropriate field mappings. This is followed by manual rewriting of the rule, iterative debugging of syntax and semantics, and repeated validation attempts directly within the target SIEM environment. Depending on rule complexity and platform differences, this process may take several hours per rule and requires deep platform-specific expertise.

In contrast, the proposed automated translation system eliminates most of these manual steps. Given a source rule in SPL or KQL, the system performs context-aware translation using Retrieval-Augmented Generation, followed by syntax validation through SIEM-native APIs and an iterative auto-fix loop. As a result, the end-to-end translation and syntax validation workflow is completed within minutes, allowing the security analyst to focus exclusively on functional validation. Moreover, this automation scales efficiently: multiple translations can be processed in parallel without requiring additional personnel.

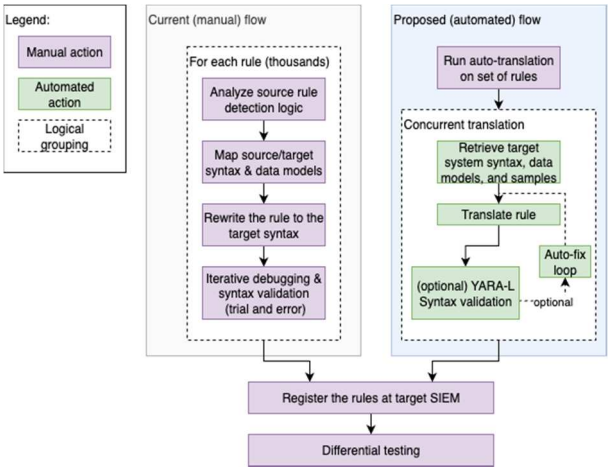


Fig. 1. Motivation Example

III. System Design

In this work, we design and implement an automated system for translating detection rules across different SIEM platforms, with a focus on reliability, transparency, and operational readiness.

At a high level, the architecture accepts a source detection rule expressed in SPL, KQL, or YARA-L and processes it through a multi-stage translation pipeline. The first stage as illustrated in Fig. 2, performs context retrieval backed by a vector store implemented with ChromaDB. This layer indexes heterogeneous datasets, including existing detection rules, platform-specific documentation, and Unified Data Model (UDM) references. Documents are chunked with context preservation and embedded using a transformer-based embedding model, enabling efficient semantic retrieval.

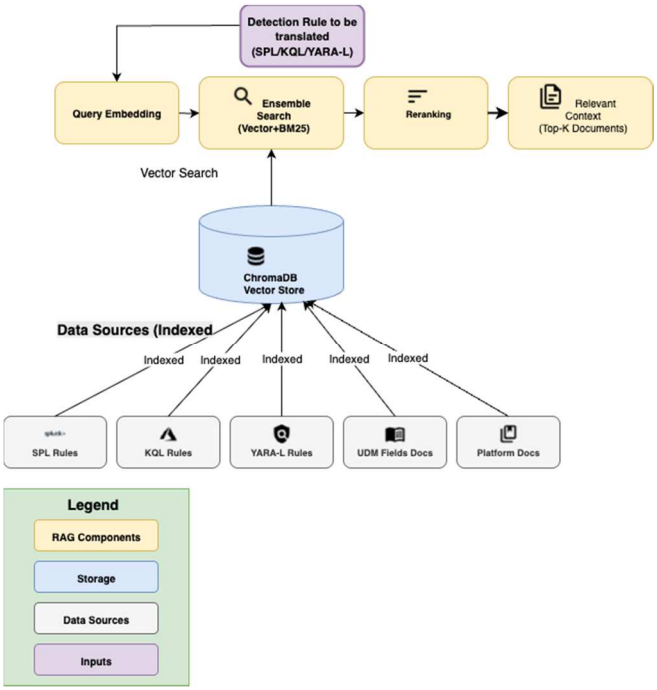


Fig. 2. Retrieval Layer

To address the limitations of purely semantic similarity, the retrieval layer employs a weighted hybrid ensemble strategy [2] combining dense vector similarity and BM25 keyword-based retrieval. Retrieved candidates are further refined using a cross-encoder re-ranking model(cross-encoder/ms-marco-MiniLM-L-6-v2), which evaluates relevance at the query–document pair level and selects a small set of high-precision examples for prompt construction.

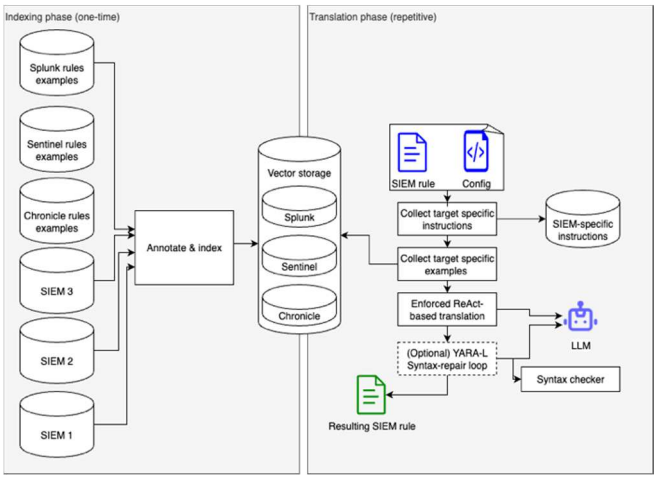


Fig. 3. System Design of Automated SIEM Detection Rule Translation System

Figure 3 illustrates the overall system design of the proposed Automated SIEM Detection Rule Translation System**.** The architecture depicts a translation pipeline in which a source detection rule expressed in SPL, KQL, or YARA-L is processed through retrieval, translation, validation, and correction stages. The retrieved context is then consumed by the translation stage, which is orchestrated using LangGraph to enforce deterministic execution flow and explicit state transitions between translation, validation, and auto-correction steps.

Figure 3 further highlights the role of SIEM-native validation APIs in verifying syntactic correctness and

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supported constructs of the generated rules. When validation errors are detected, the workflow enters an iterative auto-correction loop, as depicted in the figure, incrementally repairing the rule until it satisfies platform-specific requirements.

Workflow state is propagated across nodes using an explicit typed state representation that captures the source rule, retrieved context, translated output, validation results, correction history, and execution metadata. This stateful design enables reproducibility, auditability, and seamless integration with downstream Detection-as-Code pipelines.

IV. Conclusion

The proposed architecture builds on a successful industrial deployment at EPAM Systems [3], where an earlier version of the system was used to translate over 300 production detection use cases from KQL and SPL into YARA-L as part of a large-scale SIEM migration to Google SecOps. This deployment demonstrated that LLM-assisted rule translation can substantially reduce migration timelines while preserving detection coverage. At the same time, operational experience revealed limitations of one-shot semantic generation, including unstable field mappings and increased analyst effort for validation. The system presented in this work addresses these gaps by introducing hybrid retrieval, deterministic workflow orchestration, and native SIEM API-level validation, evolving the EPAM Systems solution into a robust, production-ready detection-as-code pipeline.

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